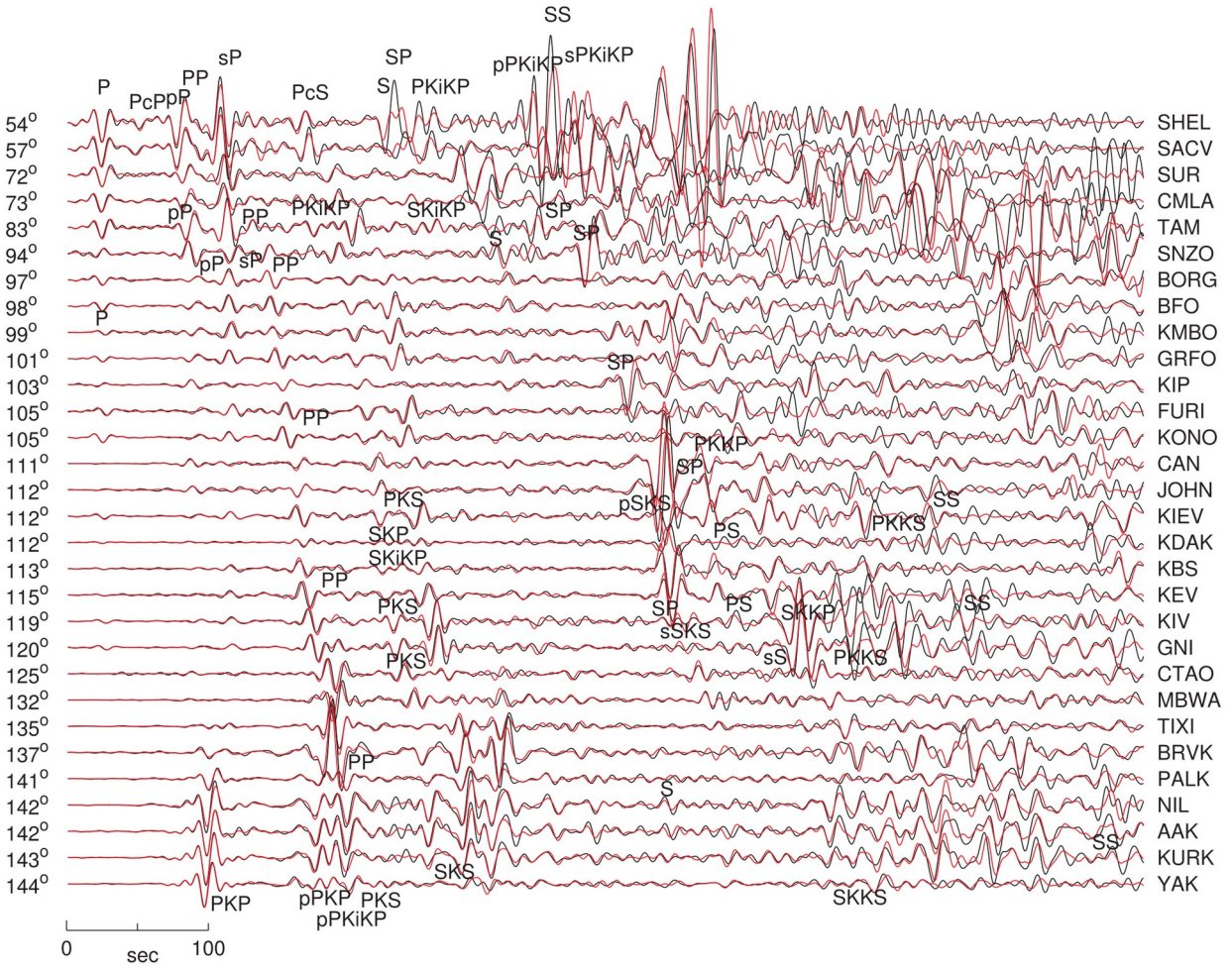
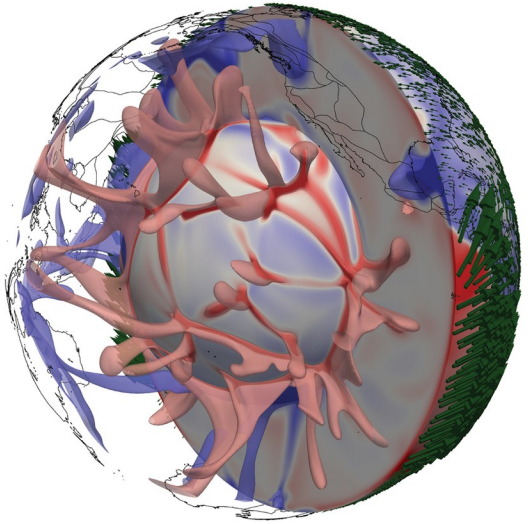
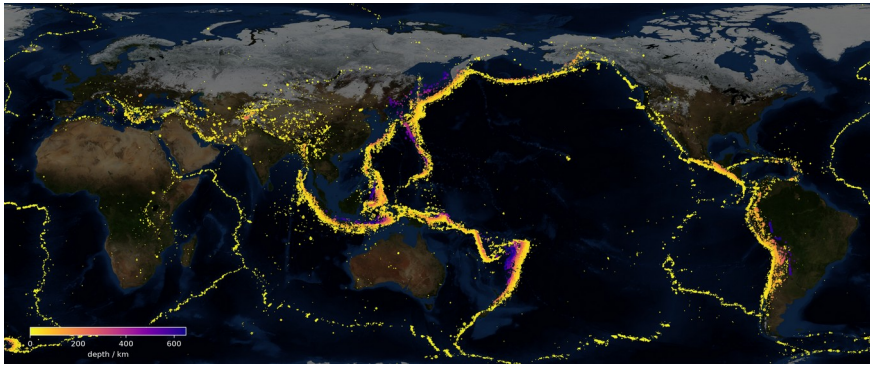
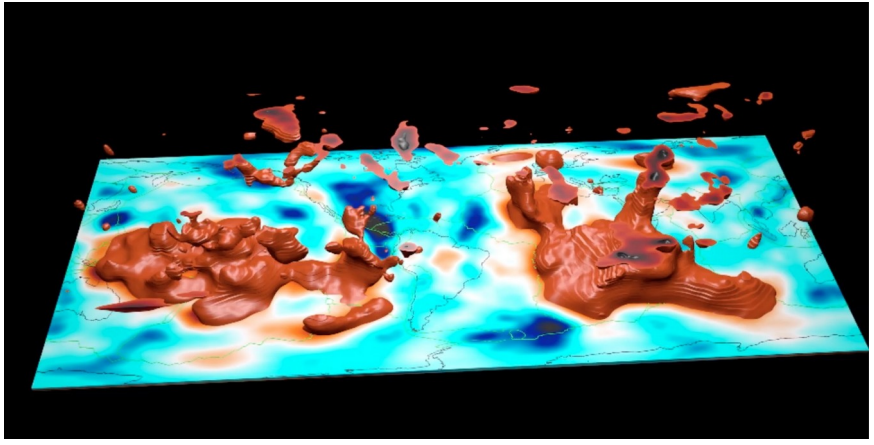
Finite elasticity

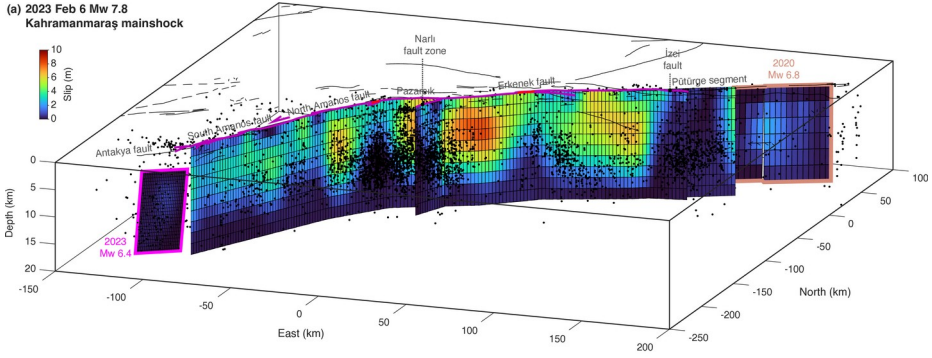


Tromp et al. 2010



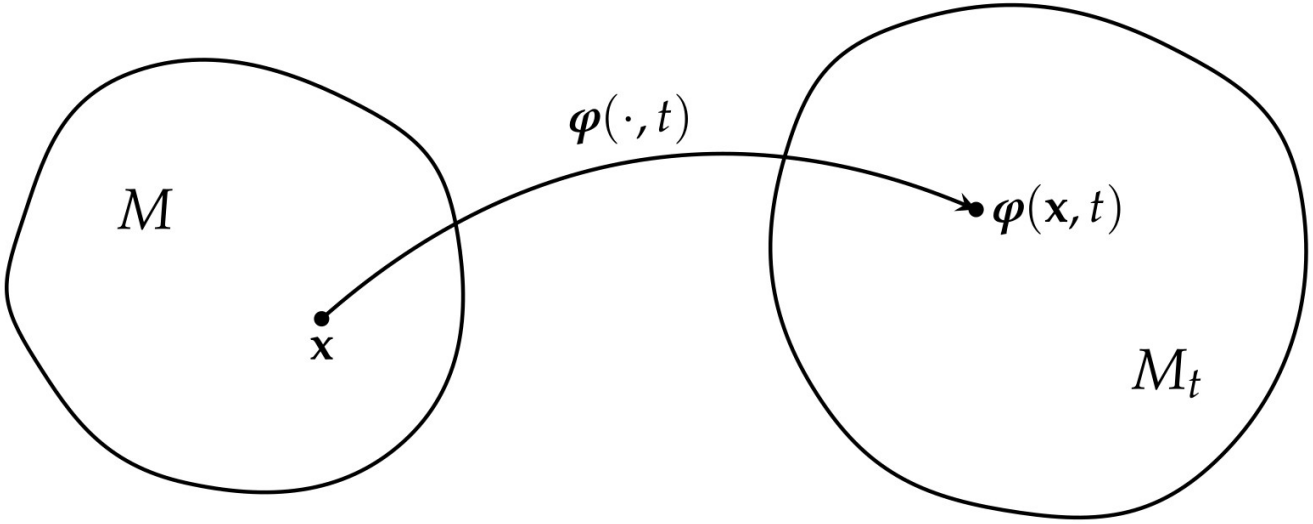






Davies *et al.* (2022); Pugmire (2025); USGS (2025); Barbot *et al.* (2023)

Kinematics



Deformation of a continuum defined relative to a reference body.

The **motion** is a field that specifies the position of particle ***x*** and time *t* in physical space

The motion is assumed to be smooth and have a smooth inverse – not fracturing or tearing or interpenetration

The conventional choice of reference body is based on a reference time:

$$\boldsymbol{\varphi}(\mathbf{x}, t_0) = \mathbf{x},$$

The (referential) **velocity** is the time derivative of the motion:

$$v_i(\mathbf{x}, t) = \frac{\partial \varphi_i}{\partial t}(\mathbf{x}, t).$$

The **deformation gradient** is the spatial derivative of the motion:

$$F_{ij} = \frac{\partial \varphi_i}{\partial x_j}.$$

$$\varphi_i(\mathbf{x} + \delta\mathbf{x}, t) = \varphi_i(\mathbf{x}, t) + F_{ij}(\mathbf{x}, t)\delta x_j + \cdots,$$

Because the motion maps the reference body one-to-one with the instantaneous body, the deformation gradient is at each point an invertible matrix. We are free to assume its determinant is **everywhere positive**.

The **Jacobian** of the motion is defined by:

$$J(\mathbf{x}, t) = \det \mathbf{F}(\mathbf{x}, t).$$

We can use it to change variables within volume integrals:

$$I = \int_{M_t} f(\mathbf{y}) \, \mathrm{d}^3\mathbf{y}.$$

$$I = \int_{M} f[\boldsymbol{\varphi}(\mathbf{x}, t)] J(\mathbf{x}, t) \, \mathrm{d}^3\mathbf{x}.$$

In this manner, we remove the need to explicitly consider moving boundaries.

Mass of a sub-body does not change with time:

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_{U_t}\varrho(\mathbf{y},t)\,\mathrm{d}^3\mathbf{y}=0.$$

Transforming this to the reference body, we get:

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_{U}J(\mathbf{x},t)\varrho[\boldsymbol{\varphi}(\mathbf{x},t),t]\,\mathrm{d}^3\mathbf{x}=0.$$

The reference volume is arbitrary, and so the integrand does not depend on time:

$$\rho(\mathbf{x})=J(\mathbf{x},t)\varrho[\boldsymbol{\varphi}(\mathbf{x},t),t].$$

The **referential density** equals the density at the reference time.

Dynamics

Kinetic energy density per unit volume in the reference body:

$$\frac{1}{2}\rho(\mathbf{x})\|\mathbf{v}(\mathbf{x},t)\|^2,$$

Total kinetic energy associated with a motion:

$$\mathcal{T}(t) = \frac{1}{2}\int_M \rho(\mathbf{x})\|\mathbf{v}(\mathbf{x},t)\|^2 \,\mathrm{d}^3\mathbf{x}.$$

This is just generalising particle mechanics to a continuum. What about the potential energy?

As a starting point, we assume that the potential energy can be written:

$$\mathcal{V}(t) = \int_M W[\mathbf{x}, t, \boldsymbol{\varphi}(\mathbf{x}, t), \mathbf{F}(\mathbf{x}, t)] \, \mathrm{d}^3\mathbf{x},$$

This choice defines a **simple hyperelastic material**

- Hyperelastic – A potential energy for the body exists (c.f., conservative forces in particle mechanics)
- Simple – The potential energy density depends only on first spatial derivatives of the motion

The key physical intuition is that elastic materials have a preferred shape or volume, and that energy is required to move them into a different configuration.

Material frame indifference requires the potential energy be invariant under a set of transformations:

Time-translation:

$$\boldsymbol{\varphi}'(\mathbf{x},t) = \boldsymbol{\varphi}(\mathbf{x},t-\tau) \longrightarrow \mathcal{V}'(t) = \mathcal{V}(t-\tau).$$

$$\mathcal{V}(t) = \int_M W[\mathbf{x}, \boldsymbol{\varphi}(\mathbf{x},t), \mathbf{F}(\mathbf{x},t)]\, \mathrm{d}^3\mathbf{x}.$$

Rigid-body motions:

$$\boldsymbol{\varphi}'(\mathbf{x},t) = \mathbf{a}(t) + \mathbf{Q}(t)\boldsymbol{\varphi}(\mathbf{x},t) \longrightarrow \mathcal{V}'(t) = \mathcal{V}(t-\tau).$$

$$W[\mathbf{x}, \mathbf{F}(\mathbf{x},t)] = W[\mathbf{x}, \mathbf{Q}(t)\mathbf{F}(\mathbf{x},t)],$$

Invariance under superimposed rigid rotations discussed further next time

The **Lagrangian density** for a simple hyperelastic body is defined by:

$$\mathcal{L}(\mathbf{x}, t, \boldsymbol{\varphi}, \mathbf{v}, \mathbf{F}) = \frac{1}{2}\rho(\mathbf{x})\|\mathbf{v}\|^2 - W(\mathbf{x}, \mathbf{F}),$$

The corresponding **action** is given by:

$$\mathcal{S}(\boldsymbol{\varphi}) = \int_0^T \int_M \mathcal{L}[\mathbf{x}, t, \boldsymbol{\varphi}(\mathbf{x}, t), \mathbf{v}(\mathbf{x}, t), \mathbf{F}(\mathbf{x}, t)]\, \mathrm{d}^3\mathbf{x}\, \mathrm{d}t,$$

Hamilton's principle states that the true motion is a **station point** of the action for variations subject to fixed endpoint conditions. This means that the **first variation of the action**:

$$\delta\mathcal{S} = \lim_{h\to 0} \frac{1}{h}[\mathcal{S}(\boldsymbol{\varphi} + h\,\delta\boldsymbol{\varphi}) - \mathcal{S}(\boldsymbol{\varphi})]$$

is zero for all perturbations to the motion such that:

$$\delta\boldsymbol{\varphi}(\mathbf{x}, 0) = \delta\boldsymbol{\varphi}(\mathbf{x}, T) = \mathbf{0}.$$

From Hamilton's principle, we arrive at the **Euler-Lagrange equations**:

$$\frac{\partial \mathcal{L}}{\partial \varphi_i} - \frac{\partial}{\partial t}\left(\frac{\partial \mathcal{L}}{\partial v_i}\right) - \frac{\partial}{\partial x_j}\left(\frac{\partial \mathcal{L}}{\partial F_{ij}}\right) = 0,$$

along with the natural boundary conditions:

$$\frac{\partial \mathcal{L}}{\partial F_{ij}} \hat{n}_j = 0,$$

Full details of the derivation are given in the handout

Using the concrete form for the Lagrangian density, the equations of motion are:

$$\rho \frac{\partial v_i}{\partial t} - \frac{\partial T_{ij}}{\partial x_j} = 0,$$

Along with the traction-free boundary conditions:

$$T_{ij} \hat{n}_j = 0.$$

Where we have defined the **first Piola-Kirchhoff** stress by:

$$T_{ij}(\mathbf{x}, t) = \frac{\partial W}{\partial F_{ij}}[\mathbf{x}, \mathbf{F}(\mathbf{x}, t)],$$

This tensor is the referential counterpart of the **Cauchy stress** seen within spatial (Eulerian) formulations of continuum mechanics.

In a referential (Lagrangian) framework, conservation of mass is trivially met, while the equations of motion are posed on a fixed reference body.